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N-1-1 Contingency-Constrained Optimal Power Flow by Interdiction Methods

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***Abstract*—*N*-1-1 contingency analysis considers the consecutive loss of two elements in a power system, with intervening time for operator adjustments; the associated reliability criterion was recently included in the NERC Standard TPL-001-1. In this paper, we introduce optimization models for *N*-1-1 contingency analysis, based on DC optimal power flow considerations. We use mixed-integer programming approaches to optimally model the system adjustments required to avoid potential cascading outages during the primary and secondary contingencies. Contingencies are determined via worst-case interdiction analysis. To facilitate operation during the secondary contingency, line overloads and load shedding are allowed. We test our models and algorithms on several IEEE test systems. Our computational experiments indicate potential for the models to augment comprehensive system operations models, such as unit commitment.**

***Index Terms*—*N*-1-1 contingency analysis, optimal power flow, power flow interdiction, mixed integer programming.**

I. Introduction

THE North American Electric Reliability Corporation (NERC) develops and enforces standards to ensure the reliability of power systems in North America. The Transmission Planning Standard (TPL-001-1) [1] defines system performance requirements under both normal and various contingency conditions, delineated as follows:

- No contingencies (Category A);
- Events resulting in the loss of a single system element (Category B);
- Events resulting in loss of two or more system elements (Category C);
- Extreme events involving loss of two or more elements, either by removal or cascading failure (Category D).

All transmission planners and planning coordinators in North America are required to prepare necessary performance assessments for their portion of the system under these conditions, and ensure that the system complies with these approved NERC TPL standards.

Category B events represent single-element contingencies, and are addressed via *N*-1 reliability analysis. A Category B event can involve the loss of a generator (B1), a transmission circuit (B2), a transformer (B3), or a single pole (DC) line (B4). *N*-1 reliability analysis has been considered in the context of unit commitment and transmission switching problems (e.g., see [2], [3]), where mixed-integer optimization models are leveraged to ensure *N*-1 compliant system operation.

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Category C events represent multiple-element contingencies. If the losses are (near-) simultaneous, the event is referred to as an *N-k* ($k \geq 2$) contingency. *N-k* contingency analysis can address a larger range of power system vulnerabilities than *N*-1 analysis. However, the combinatorial number of contingency states poses a substantial computational burden. To mitigate this burden, contingency selection is often employed. As briefly reviewed in [4], [5], approaches include online security analysis, ranking by approximate power flow solutions, concentric selection, sparse vector methods, genetic and evolutionary methods, and hybrid methods. Recently, probability analysis [6], limitations on generation and line capacity [7], swarm optimization [8], and methods based on topological characteristics of power grids [9] have been used for vulnerability analysis. Most recently, mixed-integer programming methods have been applied to *N-k* vulnerability analysis problems (see [10], [11], [12]).

A more restrictive, yet practically important alternative to *N-k* contingency analysis involves so-called *N*-1-1 contingencies. Such contingencies belong to the sub-category of events known as C3 in the TPL-001-1 [13], [1]. C3 events include a single-line-ground (SLG) or 3-phase (3Ø) fault (with normal clearing) followed by manual system adjustments, which is in turn followed by another SLG or 3Ø fault. More precisely, C3 events can be considered as two consecutive Category B events with an intervening period between them during which manual system adjustments are performed. Some simulation tools provide capabilities for *N*-1-1 analysis, e.g., see [13], [14], [15]. [16] considers an *N*-1-1 contingency analysis for the Midwest ISO.

In this paper, we introduce a computationally efficient mixed-integer optimization model for performing *N*-1-1 contingency analysis. We assume that a *N*-1-1 contingency involves the loss of a single bus (i.e., a generator, a transformer, or a substation) or transmission line, followed system adjustments (e.g., generator output level and phase angle adjustments). Following these adjustments, the system experiences subsequent loss of a single, additional bus or line. Following the definitions in [14], the *Base Case* refers to "the power system in its normal steady-state operation, with all elements in service that are expected to be in service". The first loss of a bus or a line is referred to as the *Primary Contingency*, while the second loss is referred to as the *Secondary Contingency*.

Our approach is based on an interdiction analysis of a power system (e.g., see [17]). Interdiction analysis involves identification of a subset of system components which, if interdicted (i.e., disabled), would maximize (in the worst case) the system

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loss-of-load. Two conceptual roles are played by agents in an interdiction analysis: an adversary who uses limited resources to maximize system disruption, and a system operator who manually performs system adjustments to minimize the impact of failures once they do occur. In our optimization model, both primary and secondary contingencies are selected via interdiction analysis, in which we assume the disruptive agent can disable a single element in each contingency state.

We assume the allowed system adjustments between the primary and secondary contingencies are restricted to generator output levels, phase angles, and power flows. While load shedding is not allowed during the primary contingency, the system is allowed to shed load during the secondary contingency. Further, overloads of line capacity up to 25% higher than nominal limits are allowed. Load shedding during the secondary contingency is minimized, via the introduction of a per-unit penalty cost.

Our optimization model is intended to eventually serve as a basis for advanced contingency analysis methods in comprehensive system operations models such as unit commitment. Consequently, we focus on the issue of primary and secondary contingency selection, and the linkage between the various operational states within a constrained time window (e.g., on the order of an hour - the time increment in a typical unit commitment analysis). However, we do include notional aspects of unit commitment in our optimization model, to enhance model realism and provide a basis for future integration.

The remainder of this paper is organized as follows. In Section II, we present our mixed-integer programming optimization models and associated solution algorithms for N-1-1 contingency analysis via power flow interdiction. In Section III, we analyze the performance of our approach on several IEEE test systems. Finally, we conclude in Section IV with a summary of our results.

II. Optimization Model for N-1-1 Contingency-Constrained DC Optimal Power Flow

A. Nomenclature

Index Sets and Indices

- I: The set of buses, including generators, transformers, and substations (indexed by i, j).
- $(i, j) \in L$, such that $i < j$: The set of transmission lines, $i, j \in I$. Lines are arbitrarily oriented.
- $N = (I, L)$: An undirected network representing a power system, defined by the set of buses I and the set of transmission lines L.

Parameters

- D_i: The load at bus $i \in I$. If there is no demand at bus i, $D_i = 0$.
- $\overline{F}_{ij}$: The nominal flow capacity of transmission line $(i, j) \in L$.
- $\underline{G}_i, \overline{G}_i$: Generation lower/upper bounds at bus $i \in I$, if units are committed. If there is no generation at bus i, $\underline{G}_i = \overline{G}_i = 0$.
- b_{ij}: The susceptance of line $(i, j) \in L$.
- h_i: The generation cost at bus $i \in I$.
- r_i: The penalty cost for load-shedding at bus $i \in I$.
- c_i: The commitment cost (e.g., startup and shutdown cost) for generation at bus $i \in I$.
- o_l: The allowable overload percentage of transmission line capacity during a secondary contingency; $o_l \leq 25\%$.

Decision variables

- g_i: The power generated at bus $i \in I$.
- f_{ij}: The power flow on line $(i, j) \in L$. The value is positive if power is flowing from i to j, and negative if flowing from j to i.
- s_i: The quantity of load shed at bus $i \in I$.
- θ_i: The phase angle at bus $i \in I$.
- δ_i: The interdiction indicator for bus $i \in I$. If bus i is interdicted, $\delta_i = 1$; otherwise, $\delta_i = 0$.
- σ_{ij}: The interdiction indicator for line $(i, j) \in L$. If line (i, j) is interdicted, $\sigma_{ij} = 1$; otherwise, $\sigma_{ij} = 0$.
- x_i: The unit commitment indicator for generation at bus $i \in I$. if $x_i = 1$, generation units at i are committed; otherwise, $x_i = 0$.
- $g, f, s, \theta, \delta, \sigma$: Aggregated vectors for the above-defined variables.
- The superscripts $^{(0)}$,$^{(1)}$, and $^{(2)}$ on the aggregated variables g, f, s, θ, δ, and σ denote the corresponding values during the base case, the primary contingency, and the secondary contingency scenarios, respectively.

B. Mixed-Integer Optimization Models

We now introduce three mixed-integer mathematical programming models, corresponding to the base case, primary contingency, and secondary contingency scenarios. For computational tractability, all models incorporate DC optimal power flow. The optimization objective is to minimize the total generation and load shedding penalty cost (if any); recall that load shedding is only allowed during a secondary contingency state. Following [17], we assume generation cost is linearly proportional to power output.

1) Base Case: During normal system operation, there are no failures of any system elements and is therefore no load shedding. Consequently, the base case **(BC)** optimization model reduces to a variant of security-constrained economic dispatch [19]. We have augmented the basic security-constrained economic dispatch optimization model with generation unit commitment variables, in order to (1) provide a mechanism for continuity of generator on/off state over the base case, primary contingency, and secondary contingency scenarios and (2) to facilitate eventual integration into a full-scale unit commitment model. Here, we focus on a single time period, i.e., peak demand hour, and ignore ramping constraints that in practice would be imposed on generation output level adjustments across these scenarios. The simplified base case optimization model is given as follows:

$$(\mathbf{BC}) \quad \min_{g,f,\theta,x} \quad \sum_{i\in I}(h_i g_i + c_i x_i) \tag{1}$$

$$s.t. \begin{cases} f_{ij} = b_{ij}(\theta_i - \theta_j), \forall (i,j) \in L \\ -\overline{F}_{ij} \le f_{ij} \le \overline{F}_{ij}, \forall (i,j) \in L \\ \sum_j f_{ji} + g_i = \sum_j f_{ij} + D_i, \forall i \in I \\ \underline{G}_i x_i \le g_i \le \overline{G}_i x_i, \forall i \in I \end{cases} \tag{2}$$

The optimization objective (1) is to minimize the sum of generation and commitment costs. The constraints (2) are standard DC optimal power flow constraints, which (in order): approximate active power flow on transmission lines; limit power flows on transmission lines; impose power balance at each bus; and limit generator output if generation units are committed. The optimal solution $g^{(0)}, f^{(0)}, \theta^{(0)}, x^{(0)}$ for g, f, θ, x to this model provides control parameter values for system operators under nominal conditions.

2) Primary Contingency and Operation: Next, interdiction analysis is performed to identify a worst-case scenario for the primary contingency; the analysis also yields operations parameters for the system following the primary contingency. As indicated in Section I, the primary contingency involves the loss of a single bus or transmission line. The optimization model considers objectives of both the disruptive agent (e.g., nature) and the system operator, and is consequently formulated as a max-min expression (the disruptive agent maximizes; the system operator minimizes). Additionally, we assume our test systems are capable of providing sufficient generation capacity such that no load shedding is necessary in the primary contingency, and no line overloads are required to deliver power.

Our optimization model for the primary contingency **(PC)** scenario (including subsequent operation) is as follows:

$$(\mathbf{PC}) \max_{\delta,\sigma} \min_{g,f,\theta} \quad \sum_{i\in I} h_i g_i \tag{3}$$

$$s.t. \begin{cases} f_{ij} = b_{ij}(\theta_i - \theta_j)(1-\sigma_{ij}) \\ \qquad \times (1-\delta_i)(1-\delta_j), \forall (i,j) \in L \\ -\overline{F}_{ij} \le f_{ij} \le \overline{F}_{ij}, \forall (i,j) \in L \\ \sum_j f_{ji} + g_i = \sum_j f_{ij} + D_i, \forall i \in I \\ \underline{G}_i (1-\delta_i) x_i^{(0)} \le g_i \le \overline{G}_i (1-\delta_i) x_i^{(0)}, \forall i \in I \\ \sum_{i\in I} \delta_i + \sum_{(i,j)\in L} \sigma_{ij} = 1 \\ \delta_i, \sigma_{ij} \in \{0,1\}, \forall i \in I, (i,j) \in L \end{cases} \tag{4}$$

The constraints (4) represent an interdiction variant of DC optimal power flow, which (in order): approximate active power flow on non-interdicted transmission lines (power flow across a line is zero if either terminal bus is interdicted, or the line itself is interdicted); limit power flows on transmission lines; impose power balance at each bus; limit generator output at non-interdicted buses, if units are committed; ensure that only a single network element is interdicted. Different from [17], we do not consider the multiplicative term $(1-\sigma_{ij})(1-\delta_i)(1-\delta_j)$ on the line flow limit constraint; its inclusion in the constraint for approximating active power flow is sufficient.

Optimal solutions $\delta^{(1)}, \sigma^{(1)}, g^{(1)}, f^{(1)}, \theta^{(1)}$ provide the worst-case interdicted bus or line (by $\delta^{(1)}$ and $\sigma^{(1)}$) in the primary contingency scenario, in addition to the minimal-cost system adjustments $g^{(1)}, f^{(1)}, \theta^{(1)}$ to be performed following the contingency.

3) Secondary Contingency and Operations: After the primary contingency and manual system adjustments, in a secondary contingency follows. Here, we again use the interdiction method to select the worst-case secondary contingency, and to identify the optimal system adjustments after the secondary contingency. In a secondary contingency, load shedding and line overloads are permitted for emergency control and to regain power system stability. To return to normal system operation, a restoration adjustment is needed. The model for secondary contingency **(SC)** selection and system operation is presented in (5) and (6).

In this model, we assume that the disruptive agents in the primary contingency and secondary contingency are independent. Therefore, in the model **(SC)**, the first interdiction results $\delta^{(1)}, \sigma^{(1)}$ are known. This model is quite similar to **(PC)** but the penalty cost $\sum_i r_i s_i$ for load shedding is included in the objective, and real consumption at bus i is $D_i - s_i$. The load shedding amount at a particular bus is upper bounded by the bus demand. Additionally, the line capacity is allowed to have overloads up to a factor of $1 + o_l$. At the same time, the bus or line lost in the primary contingency will not be the target of the second disruptive agent; this is guaranteed by $\delta_i \times \delta_i^{(1)} = 0, \sigma_{ij} \times \sigma_{ij}^{(1)} = 0$.

Again, by solving this program, the optimal solutions $\delta^{(2)}, \sigma^{(2)}$ will identify the worst-case bus or line outage in a secondary contingency. In the meantime, the solution of **(SC)** also identifies emergency operation adjustments after the secondary contingency. This is given by the optimal **(SC)** solutions $g^{(2)}, f^{(2)}, s^{(2)}, \theta^{(2)}$.

C. Algorithms for the N-1-1 Contingency Analysis

Both the models **(PC)** and **(SC)** are mixed integer nonlinear programs with max-min objectives. We follow reformulation methods in [12] by dualizing the inner minimization problem and then applying linearization techniques to obtain equivalent mixed integer linear programs.

By fixing δ, σ in the model **(PC)**, dualizing the inner minimization problem (which has a finite optimal solution) and then combining the unknown variables δ, σ, we derived an equivalent formulation **(PC)**$'$ (7)-(13) in place of **(PC)**. Parameters on the right-hand side denote corresponding primal variables. Similarly, for model **(SC)**, we derive an equivalent formulation **(SC)**$'$ given by (14)-(23).

Observe that in formulation (7)-(13), the terms: $\alpha_{ij}(1-\delta_i)(1-\delta_j)(1-\sigma_{ij}), (1-\delta_i)\zeta_i, (1-\delta_i)\xi_i$ where $i = 1, \cdots, I, j = i+1, \cdots, I$, are nonlinear. To linearize them, we introduce additional continuous variables: $u_{ij} = (1-\delta_i)(1-\delta_j)(1-\sigma_{ij})$, $v_{ij} = \alpha_{ij} u_{ij}$, $wa_i = (1-\delta_i)\zeta_i$ and $wb_i = (1-\delta_i)\xi_i$. The term

$$(\mathbf{SC}) \quad \max_{\delta,\sigma} \min_{g,s,f,\theta} \quad \sum_{i\in I}(h_i g_i + r_i s_i) \tag{5}$$

$$s.t. \begin{cases} f_{ij} = b_{ij}(\theta_i - \theta_j)[(1-\delta_i^{(1)})(1-\delta_j^{(1)})(1-\sigma_{ij}^{(1)})][(1-\delta_i)(1-\delta_j)(1-\sigma_{ij})], \forall (i,j) \in L \\ -\overline{F}_{ij}(1-\sigma_{ij}^{(1)})(1+o_l) \le f_{ij} \le \overline{F}_{ij}(1-\sigma_{ij}^{(1)})(1+o_l), \forall (i,j) \in L \\ \sum_j f_{ji} + g_i = \sum_j f_{ij} + (D_i - s_i), \forall i \in I \\ \underline{G}_i(1-\delta_i^{(1)})(1-\delta_i)x_i^{(0)} \le g_i \le \overline{G}_i(1-\delta_i^{(1)})(1-\delta_i)x_i^{(0)}, \forall i \in I \\ 0 \le s_i \le D_i, \forall i \in I \\ \sum_{i\in I}\delta_i + \sum_{(i,j)\in L}\sigma_{ij} = 1 \\ \delta_i, \sigma_{ij} \in \{0,1\}, \forall i \in I, (i,j) \in L \\ \delta_i \times \delta_i^{(1)} = 0, \forall i \in I \\ \sigma_{ij} \times \sigma_{ij}^{(1)} = 0, \forall (i,j) \in L \end{cases} \tag{6}$$

$$(\mathbf{PC})' \quad \max_{\delta,\sigma,\alpha,\beta,\gamma,\eta,\zeta,\xi} \sum_{(i,j)\in L} \overline{F}_{ij}(-\beta_{ij} + \gamma_{ij}) + \sum_{i\in I}(D_i\eta_i + \overline{G}_i x_i^{(0)}(1-\delta_i)\zeta_i + \underline{G}_i x_i^{(0)}(1-\delta_i)\xi_i) \tag{7}$$

$$s.t. \quad \eta_i + \zeta_i + \xi_i \le h_i \qquad \leftarrow g_i \tag{8}$$

$$\alpha_{ij} + \beta_{ij} + \gamma_{ij} - \eta_i + \eta_j = 0 \qquad \leftarrow f_{ij} \tag{9}$$

$$-\sum_{j:j>i} b_{ij}(1-\delta_i)(1-\delta_j)(1-\sigma_{ij})\alpha_{ij} + \sum_{j:j<i} b_{ji}(1-\delta_j)(1-\delta_i)(1-\sigma_{ji})\alpha_{ji} = 0 \qquad \leftarrow \theta_i \tag{10}$$

$$\beta_{ij} \ge 0, \gamma_{ij} \le 0, \zeta_i \le 0, \xi_i \ge 0 \tag{11}$$

$$\sum_{i\in I}\delta_i + \sum_{(i,j)\in L}\sigma_{ij} = 1 \tag{12}$$

$$\delta_i, \sigma_{ij} \in \{0,1\}, \forall i \in I, (i,j) \in L \tag{13}$$

$$(\mathbf{SC})' \quad \max_{\delta,\sigma,\alpha,\beta,\gamma,\eta,\zeta,\mu} \sum_{(i,j)\in L} \overline{F}_{ij}(1-\sigma_{ij}^{(1)})(1+o_l)(-\beta_{ij} + \gamma_{ij}) \tag{14}$$

$$+\sum_{i\in I}[D_i(\eta_i + \mu_i) + \overline{G}_i x_i^{(0)}(1-\delta_i^{(1)})(1-\delta_i)\zeta_i + \underline{G}_i x_i^{(0)}(1-\delta_i^{(1)})(1-\delta_i)\xi_i]$$

$$s.t. \quad \eta_i + \zeta_i + \xi_i \le h_i \qquad \leftarrow g_i \tag{15}$$

$$\eta_i + \mu_i \le r_i \qquad \leftarrow s_i \tag{16}$$

$$\alpha_{ij} + \beta_{ij} + \gamma_{ij} - \eta_i + \eta_j = 0 \qquad \leftarrow f_{ij} \tag{17}$$

$$-\sum_{j:j>i} b_{ij}[(1-\delta_i^{(1)})(1-\delta_j^{(1)})(1-\sigma_{ij}^{(1)})][(1-\delta_i)(1-\delta_j)(1-\sigma_{ij})]\alpha_{ij}$$

$$+\sum_{j:j<i} b_{ji}[(1-\delta_j^{(1)})(1-\delta_i^{(1)})(1-\sigma_{ji}^{(1)})][(1-\delta_j)(1-\delta_i)(1-\sigma_{ji})]\alpha_{ji} = 0 \qquad \leftarrow \theta_i \tag{18}$$

$$\beta_{ij} \ge 0, \gamma_{ij} \le 0, \zeta_i \le 0, \xi_i \ge 0, \mu_i \le 0 \tag{19}$$

$$\sum_{i\in I}\delta_i + \sum_{(i,j)\in L}\sigma_{ij} = 1 \tag{20}$$

$$\delta_i, \sigma_{ij} \in \{0,1\}, \forall i \in I, (i,j) \in L \tag{21}$$

$$\delta_i \times \delta_i^{(1)} = 0, \forall i \in I \tag{22}$$

$$\sigma_{ij} \times \sigma_{ij}^{(1)} = 0, \forall (i,j) \in L \tag{23}$$

$u_{ij} = (1-\delta_i)(1-\delta_j)(1-\sigma_{ij})$ can be linearized by:

$$\begin{cases} u_{ij} \le 1 - \delta_i \\ u_{ij} \le 1 - \delta_j \\ u_{ij} \le 1 - \sigma_{ij} \\ u_{ij} \ge 1 - \delta_i - \delta_j - \sigma_{ij} \\ u_{ij} \ge 0, \end{cases} \tag{24}$$

Similarly, term $v_{ij} = \alpha_{ij}u_{ij}$ can be linearized by:

$$\begin{cases} v_{ij} \ge \alpha_{ij} - Ub \cdot (1 - u_{ij}) \\ v_{ij} \ge Lb \cdot u_{ij} \\ v_{ij} \le \alpha_{ij} - Lb \cdot (1 - u_{ij}) \\ v_{ij} \le Ub \cdot u_{ij}, \end{cases} \tag{25}$$

and term $wa_i = (1-\delta_i)\zeta_i$ can be linearized by:

$$\begin{cases} wa_i \geq \zeta_i - Ub \cdot \delta_i \\ wa_i \geq Lb \cdot (1-\delta_i) \\ wa_i \leq \zeta_i - Lb \cdot \delta_i \\ wa_i \leq Ub \cdot (1-\delta_i). \end{cases} \tag{26}$$

where Lb, Ub (parameters) are given lower and upper bounds for α_{ij}, ζ_i and are assumed to be sufficiently small and large, respectively. wb_i can be linearized similarly. Using the variables $u_{ij}, v_{ij}, wa_i, wb_i$ to replace their equivalent counterparts in the formulation **(PC)**$'$ and combining (24)-(26), we obtain an equivalent mixed integer linear program (MILP) formulation for **(PC)**. We follow a similar strategy to linearize **(SC)**$'$.

Based on these equivalent formulations, we present the N-1-1 contingency analysis process as follows (see Table I).

TABLE I
N-1-1 CONTINGENCY ANALYSIS BY INTERDICTION METHODS

- Step 1: Base Case: solve the MILP formulation (1)-(2) by CPLEX for the model **(BC)** to obtain the optimal operations $g^{(0)}, f^{(0)}, \theta^{(0)}, x^{(0)}$ under normal situations;
- Step 2: Primary Contingency and Operations: solve the equivalent formulation (7)-(13) with linearizations in (24)-(26) by CPLEX for the model **(PC)**: (the original min part must be feasible and then duality can be taken)
  - The optimal solutions $\delta^{(1)}, \sigma^{(1)}$ is the possible action a disruptive agent may take for this contingency;
  - Plug the values of $\delta^{(1)}, \sigma^{(1)}$ into the **(PC)**$'$ model, and obtain the dual solutions $g^{(1)}, f^{(1)}, \theta^{(1)}$;
  - The following system adjustments performed by system operators are given by the optimal solutions $g^{(1)}, f^{(1)}, \theta^{(1)}$;
- Step 3: Secondary Contingency and Operations: solve the equivalent formulation (14)-(23) with linearizations in (24)-(26) by CPLEX for the model **(SC)**: (the original min part is always feasible)
  - The optimal solutions $\delta^{(2)}, \sigma^{(2)}$ is the possible action a disruptive agent may take for this contingency;
  - Plug the values of $\delta^{(2)}, \sigma^{(2)}$ into the **(SC)**$'$ model, and obtain the dual solutions $g^{(2)}, f^{(2)}, s^{(2)}, \theta^{(2)}$;
  - The following system adjustments for emergency operation performed by system operators are given by the optimal solutions $g^{(2)}, f^{(2)}, s^{(2)}, \theta^{(2)}$ (load shedding and overloads on lines are allowed for emergency operation to ensure reliability).

III. NUMERICAL EXPERIMENTS

Our algorithm for N-1-1 contingency analysis is implemented in C++, using CPLEX 12.1 as the mixed-integer solver. CPLEX capabilities are accessed via the CPLEX Concert 2.9 C++ library. The CPLEX optimality gap is set to 0.1%. In the following experiments, if we assume a line (i, j) cannot be interdicted, $\sigma_{ij} = 0$ is added to corresponding models; similarly, if a bus i cannot be interdicted, $\delta_i = 0$ is added.

We perform preliminary testing using a simple, 5-bus test system. This system is derived from the example described in [15], and has 3 generators and 2 load buses (see Fig. 1). The maximum output $\overline{G}_i$ for the generators at buses 1-3 are respectively 3.5MW, 2.2MW, and 2.2MW; the demands at each of the load buses 4 and 5 is 3MW. Once committed, generator minimum output levels at buses 1-3 are respectively 0.3MW, 0.3MW, and 0.3MW. All transmission lines have flow capacities of 1.8MW, and susceptances $b_{ij} = 1$. Finally, we define $h_i = c_i = 1$ for $i \in \{1,2,3\}$, $r_4 = 3$, and $r_5 = 2$.

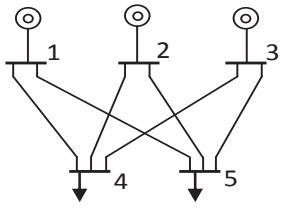


Fig. 1. An illustrative 5-bus test system

TABLE II
GENERATION DATA FOR THE 5-BUS TEST SYSTEM (UNITS: MW)

Generators			**(BC)**		**(PC)**	**(SC)**
Number	$\underline{G}_i$	$\overline{G}_i$	$x_i^{(0)}$	$g_i^{(0)}$	$g_i^{(1)}$	$g_i^{(2)}$
1	0.3	3.5	1	1.6	3.4	3.5
2	0.3	2.2	1	2.2	2.2	0.3
3	0.3	2.2	1	2.2	0.4	1.45

Assume that all interdictions are on only on lines. The results of our analysis are shown in Table II and Table III. Algorithm run-times for this case were negligible, and are not reported. The primary contingency scenario involves the loss of line $(3,5)$, such that generator output at buses 1 and 3 must be adjusted to satisfy the demand at buses 4 and 5. The secondary contingency scenario involves the loss of line $(2,5)$, leaving a single line connected to load bus 5. Even with allowable line overload in a secondary contingency (due to $o_l = 25\%$), bus 5 must shed 0.75MW of load.

Next, we consider the more comprehensive IEEE 30-Bus and RTS-96 test systems [18]. To ensure feasibility of our optimization models (which is required for dualization), we have performed minor modifications of generation and load for these systems. In the **(PC)** optimization model, consider a load bus with no local generation that is served by a single transmission line. If the incident transmission line is interdicted, the optimization problem is infeasible, as demand cannot be fully satisfied. Similarly, in both **(PC)** and **(SC)** models, if the only line connected to a generation bus is interdicted, any output (if the unit is enabled) cannot be transmitted. To ensure feasibility is such cases, we prevent the interdiction of such radial transmission lines. Radial lines are often excluded from reliability standards [20] and thus excluded from contingency analysis.

Results for the IEEE 30-bus system (with 30 buses - 6 of which have generation capabilities - and 41 transmission lines) are shown in Table IV. To avoid infeasibilities, we disallow interdiction of lines that serve as sole supply for a load bus. Run-times for solving the **(BC)**, **(PC)**, and **(SC)** optimization models are respectively 0.01, 0.07, and 0.16 seconds.

The primary contingency scenario involves the loss of line $(12,15)$, which has no impact on the generation levels. The second contingency scenario involves the loss of bus 6, which yields some load shedding and corresponding drops in generation levels. Specifically, buses 10, 18, 29, and 30 respectively shed 1.3MW, 3.2MW, 2.4MW, and 10.6MW.

TABLE III
POWER FLOWS FOR THE 5-BUS TEST SYSTEM (UNITS: MW)

Lines		(BC)	(PC)	(SC)
(i,j)	$\overline{F}_{ij}$	$f_{ij}^{(0)}$	$f_{ij}^{(1)}$	$f_{ij}^{(2)}$
(1,4)	1.8	0.8	1.6	1.25
(1,5)	1.8	0.8	1.8	2.25^{+}
(2,4)	1.8	1.1	1.0	0.30
(2,5)	1.8	1.1	1.2	-
(3,4)	1.8	1.1	0.4	1.45
(3,5)	1.8	1.1	-	-

Note: $^{+}$ / - entries respectively denote overloaded / interdicted lines.

TABLE IV
GENERATION DATA FOR THE IEEE-30-BUS SYSTEM (UNITS: MW)

Generators				(BC)		(PC)	(SC)
Number	Name	$\underline{G}_i$	$\overline{G}_i$	$x_i^{(0)}$	$g_i^{(0)}$	$g_i^{(1)}$	$g_i^{(2)}$
1	Glen Lyn	10	30	0	0	0	0
2	Claytor	10	50	1	50	50	50
5	Fieldale	10	30	0	0	0	0
8	Reusens	10	60	1	60	60	60
11	Roanoke	10	70	0	0	0	0
13	Hancock	10	60	1	27.5	27.5	10

Next, we consider results (not shown) for the RTS-96 system [18], which has 73 buses and 120 lines. We aggregate lines with identical endpoints by summing their susceptances and line capacities. We still assume that interdictions cannot be on lines that are connected to any load buses. Additionally, in **(PC)**, we assume interdictions cannot be on generation buses and load buses to ensure the feasibility of the inner minimization problem. However, in **(SC)**, any bus can be interdicted. In 0.02, 0.30, and 13.22 seconds of run-time we obtained worst-case contingencies for our N-1-1 optimization models. The primary and secondary contingencies respectively involve the loss of line $(323,325)$(C35) and bus 223(Bloch). The amount of load shed is 13% of the total demand; further, one line ($(216,219)$(B28)) is overloaded.

IV. CONCLUSIONS

The N-1-1 reliability criterion is an important alternative to the N-1 and N-k criteria, providing for more robust systems than the former and improved computational tractability relative to the latter. However, N-1-1 analysis is in its infancy, and associated computational methods are not well-established.

We have introduced an optimization method for N-1-1 contingency analysis, based on interdiction techniques. Our model requires full compliance in the primary contingency, but allows for line overloads and load shedding during the secondary contingency. Additionally, we assume that generation levels can be adjusted as necessary between the base case, primary contingency, and secondary contingency scenarios.

Computational experience with our models suggests a feasible baseline for integration with comprehensive systems operations models such as unit commitment. However, such efforts will require consideration of additional factors, e.g., adherence to generator ramp rates between the primary and secondary contingencies. Further, we presently assume that disruptive agents in the primary and secondary contingencies are independent. If these contingencies are correlated, more complex bi-level optimization models must be considered.

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